

What we're doing

Can Helico's own confidence rank native and near-native protein structures above Rosetta decoys when each candidate is converted to Helico's contact-map representation, and how does it compare with AF2Rank, DeepAccNet, and the Rosetta energy function on the exact AF2Rank benchmark?

Why

A candidate-derived contact map that is geometrically compatible with the target sequence should let contact-conditioned Helico produce a confident structure, whereas inconsistent decoy contacts should reduce Helico pTM. We therefore expect target-wise Helico pTM to correlate positively with candidate TM-score and recover high-quality candidates substantially better than Rosetta energy, but probably below AF2Rank because the contact map discards most of the template geometry available to AlphaFold. The native-versus-decoy discrimination question and the standard decoy-quality ranking question will both be reported.

Results so far

The exact public benchmark is available: 133 targets, 180,079 decoys, 133 natives, and corrected AF2Rank outputs all join without a missing key. Our analysis reproduces the paper's full-set AF2Rank / DeepAccNet / Rosetta rank correlations (0.925 / 0.831 / 0.759).

Target-balanced pilot

We ran nine targets x (24 TM-stratified decoys + native), three Helico diffusion samples each. All 225 candidates succeeded. Primary Helico pTM reaches 0.852 mean target-wise Spearman correlation with candidate TM-score, versus 0.944 for AF2Rank, 0.871 for DeepAccNet, and 0.820 for Rosetta on the paired pilot.

The secondary Helico composite (pTM times candidate/output TM-score) reaches 0.889 correlation and the best pilot top-1 TM-score, 0.958. It uses more than confidence and is reported separately.

Native versus decoy

Helico pTM ranks the exact native first on 3/9 targets and gives a mean native rank of 3.22 among 25 candidates. AF2Rank's composite is top-1 on 5/9; the Helico composite is top-1 on 7/9. Pure confidence is useful but does not by itself reliably identify the exact native.

Full CoreWeave run

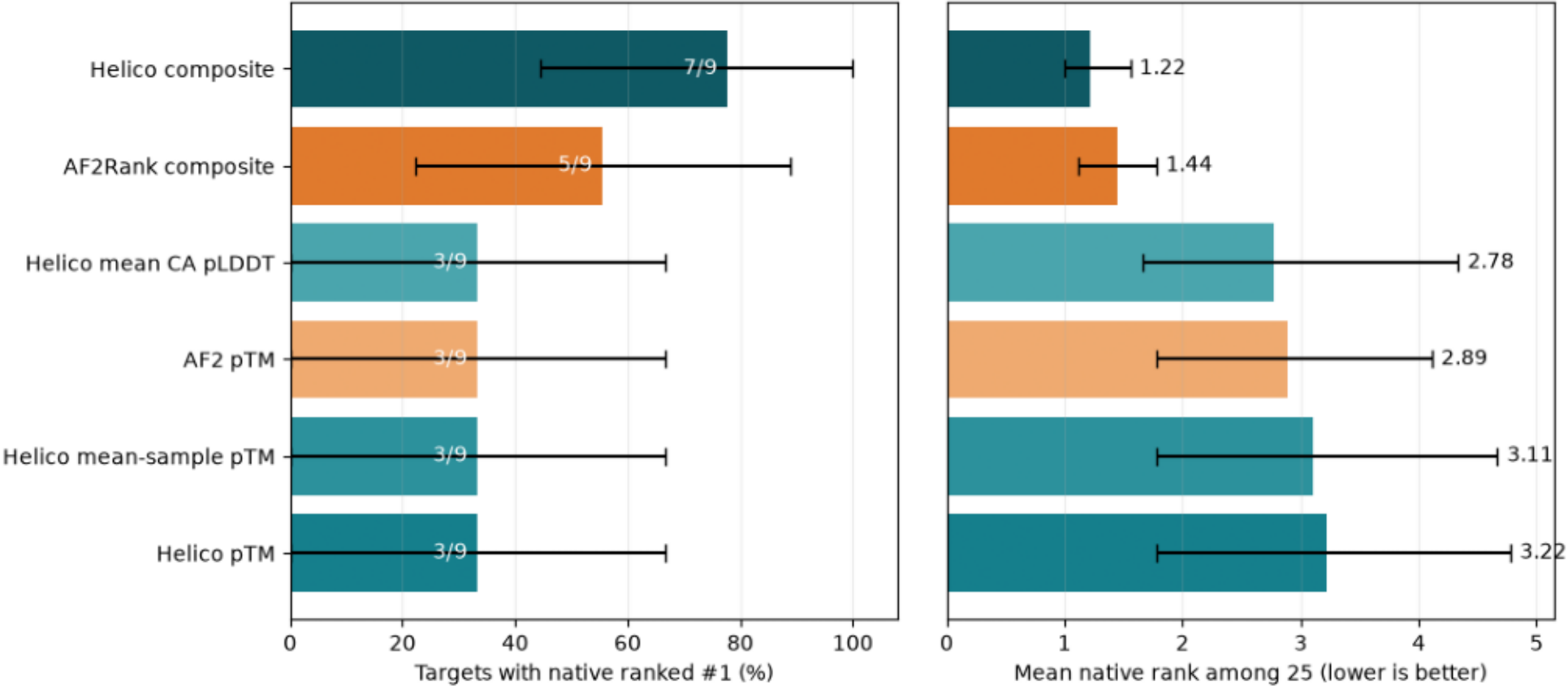
The pilot used 0.544 inference H100-hours. One-candidate-at-a-time extrapolation is about 436 H100-hours / \$1,722 for all 180,212 candidates. The complete run is being distributed across 96 independent CoreWeave H100 jobs with resumable 32-candidate parts. Workers reuse the model and target tokenization while preserving the pilot's same-seed-per-candidate protocol. Expected ideal wall time is about 4.5 inference hours, plus startup and scheduling.

Both pre-launch smokes passed, including the 223-residue maximum-length target. All 180,212 candidates are retained. A systematic extra N-terminal lysine in 1,000 `1iib` decoys is uniquely mapped and explicitly removed rather than silently shifting indices or excluding candidates.

pilot_native_selection.png

Native top-1 recovery and mean native rank across the nine-target pilot. Each target contains its native and 24 TM-stratified decoys. DeepAccNet and Rosetta cannot be scored for native recovery because their corrected AF2Rank native rows contain -1 sentinels.

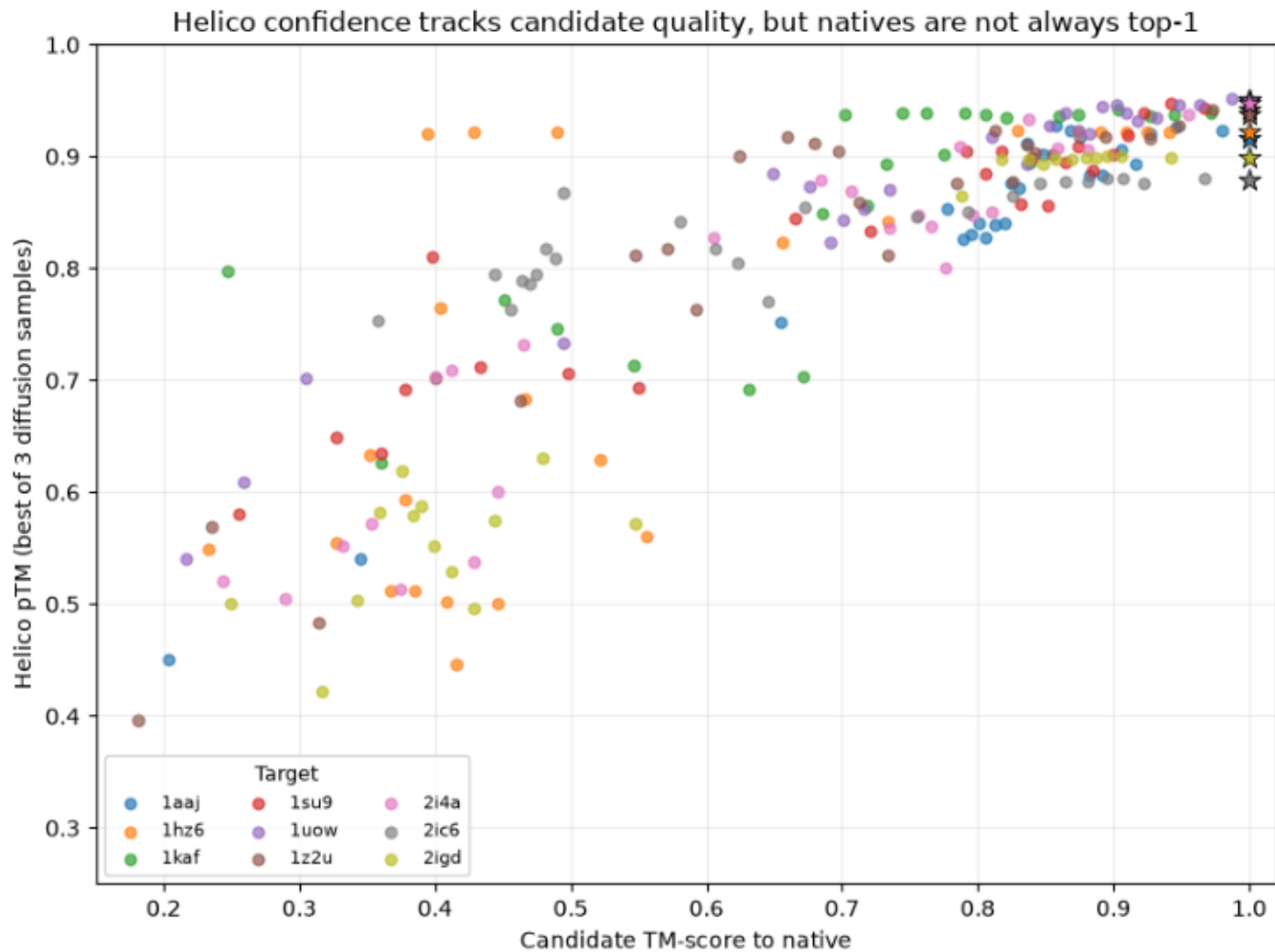
Native identification in the pilot: 9 targets, 24 decoys + native each



Error bars: 95% target-bootstrap interval. DeepAccNet and Rosetta omitted: their AF2Rank native rows contain -1 sentinel scores.

pilot_ptm_vs_tm_score.png

Each circle is a TM-stratified Rosetta decoy; stars are native structures. Colors denote the nine length/count-stratified targets.



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$ plot_pilot.py
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pilot_spearman_comparison.png

Mean target-wise Spearman correlation with reference TM-score; bars are 95% target-bootstrap intervals. Helico pTM is the primary score; the Helico composite additionally uses candidate/output TM-score.

